

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Computational Neuroscience [NPT_4130]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Describe the basic structures in the brain involved in sensory and motor functions. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Understand"
B) Explain how networks of neurons and synapses form systems of neural processing. (5)
Course Outcome: "CO1" / Cognitive Level: "Understand"
- 2) Illustrate how the patch-clamp technique is used to study ion channels and membrane potentials. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
B) Analyse how the different ionic conductances and gating variables in the Hodgkin-Huxley model interact to produce an action potential. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 3) Using the Morris-Lecar model, show how bifurcation analysis helps in understanding neural firing patterns. (5)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) Analyze how changes in external stimuli alter the trajectory patterns in the phase-plane of a neuron model, and infer what this reveals about neural excitability. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Analyse how different statistical methods influence spike train discrimination and interpret their impact on decoding sensory information. (5)
A) Course Outcome: "CO4" / Cognitive Level: "Analysis"
B) Illustrate with an example how population coding improves stimulus representation in neural systems. (5)
Course Outcome: "CO4" / Cognitive Level: "Application"
- 5) Analyse how different mechanisms of short-term plasticity influence neural computation and information processing. (5)
A)

- B) Evaluate the role of long-term potentiation (LTP) and long-term depression (LTD) in learning and memory. (5)

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Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4130:Computational Neuroscience

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	5
3	3	CO 3	15
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4130:Computational Neuroscience

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	2	10(20%)
4	Analysis	5	25(50%)
5	Create	1	5(10%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)